ASE CHAPTER 1

T5 · Evaluate

Judgement questions grounded in Aggarwal — answers in English, traps and notes in Arabic

Chapter 1 — Introduction

كيف تستخدم الملف. اقرأ السؤال، احكم بنفسك وبرّر، بعدين افتح البطاقة. **الجواب بالإنجليزي** — هذا اللي تترجمه بالورقة. **البطاقات الصفراء والحمراء بالعربي** — فخاخ وتنبيهات.

المصدر. Aggarwal & Singh, *Software Engineering* 3rd ed. (الطبعة الثالثة) — السكان AliFahim_Software_Engineering.pdf. المدى = ص.1–16 · مفتاح الأجوبة = ص.498. أغروال هو المصدر الأساسي؛ الكتب الأخرى تُستخدم فقط لنقطة مطلوبة ناقصة عنده.

طريقة قراءة الإجابات

الكلام العريض مبني على المصدر، وقد يُبسّط بألفاظ أسهل. الكلام العادي صياغة أو استنتاج مختصر. الاقتباس الحرفي فقط يوضع بين علامتي تنصيص.

Retrieval set — T5 Evaluate

E.1 · Is software engineering an art, a craft, or a science? Judge and justify your position.

Answer

Software engineering is best described as an **engineering discipline**: systematic in its principles and still dependent on design judgement. The **art of programming alone is no longer enough to build large programs**, and software engineering is an important discipline of study, practice and research.

Aggarwal, §1.2.1, p.4; §1.6.4, p.16.

E.2 · Which type of software product accounts for more development projects — generic or customised? Evaluate and justify.

Answer

Generic products are **"developed for anonymous customers"** — the target is **"the entire world"** and many copies are sold (operating systems, compilers, word processors, CASE tools). **Customised products** are **"developed for particular customers"** to their specific requirements. Judging by development volume, customised is larger: **"most of the development projects (say about 80%) come under this category."**

Aggarwal, §1.5.7, p.14.

دقّق بالوحدة قبل ما تحكم

الـ80% نسبة **من المشاريع المطوَّرة**، مو من المستخدمين ولا من النسخ. المنتج generic يُباع «للعالم كله» بنسخ كثيرة فيوصل مستخدمين أكثر.

E.3 · What is more important: product or process? Justify your judgement.

Answer

"An efficient process is required to produce good quality products. If the process is weak, the end product will undoubtedly suffer, but an obsessive over-reliance on process is also dangerous." The aim is to focus **"on both the quality of the product and on the process used to develop the product"**, and the dependency order puts process **"after people and product."**

Neither is universally more important: an effective process supports product quality, while excessive reliance on process can be harmful. The right balance depends on the product and its requirements.

Aggarwal, §1.5.2, p.13; §1.6.3, p.16; Fig. 1.8, p.15.

Rapid self-test — recall prompts (no answers shown)

Give your reasoned position on software engineering as art, craft, or science.
Which product type is the larger share of development, and what exactly is the 80% measuring?
Explain why a sound process helps product quality and why over-relying on process can be harmful.
What is the management dependency order of the four factors?